

Pandas Syllabus

1. Introduction to Pandas

- What is Pandas
- Why use Pandas in Data Analytics
- Features of Pandas
- Installing [Pandas](#)
- Importing Pandas

```
import pandas as pd
```

2. Pandas Data Structures

Series

- Creating Series
- Indexing in Series
- Operations on Series

DataFrame

- Creating DataFrame
 - Rows and Columns
 - DataFrame from:
 - List
 - Dictionary
 - CSV
 - Excel
 - SQL
-

3. Reading and Writing Data

- `read_csv()`
- `read_excel()`
- `read_json()`
- `read_sql()`
- `to_csv()`
- `to_excel()`

Example:

```
df = pd.read_csv("students.csv")
```

4. Viewing and Understanding Data

- `head()`
 - `tail()`
 - `shape`
 - `columns`
 - `info()`
 - `describe()`
 - `dtypes`
-

5. Selecting Data

- Selecting Columns
- Selecting Rows
- loc[]
- iloc[]
- Filtering Data
- Conditional Selection

Example:

```
df[df["Marks"] > 50]
```

6. Data Cleaning

- Handling Missing Values
 - isnull()
 - dropna()
 - fillna()
 - Removing Duplicates
 - Renaming Columns
 - Changing Data Types
-

7. Data Manipulation

- Adding Columns
 - Deleting Columns
 - Updating Values
 - Sorting Data
 - Value Counts
 - Unique Values
-

8. String Operations

- `str.lower()`
 - `str.upper()`
 - `str.contains()`
 - `str.replace()`
 - Splitting Strings
-

9. Mathematical Operations

- Sum
- Mean
- Median
- Mode
- Min / Max
- Standard Deviation

Example:

```
df["Salary"].mean()
```

10. Grouping and Aggregation

- `groupby()`
- Aggregate Functions
- Multiple Aggregations
- Pivot Table

Example:

```
df.groupby("Department")["Salary"].sum()
```

11. Merging and Joining

- `merge()`
 - `concat()`
 - Join Types:
 - Inner Join
 - Left Join
 - Right Join
 - Outer Join
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12. Date and Time Handling

- Working with Dates
 - Converting Date Format
 - Extracting:
 - Year
 - Month
 - Day
 - Time Series Basics
-

13. Apply Functions

- `apply()`
 - Lambda Functions
 - Map Functions
-

14. Indexing

- Setting Index
- Resetting Index
- MultiIndex

15. Advanced Pandas

- Multi-Level Grouping
- Crosstab
- Window Functions
- Reshaping Data
- Melt and Pivot

16. Visualization with Pandas

- Line Chart
- Bar Chart
- Pie Chart
- Histogram

Using:

- [Matplotlib](#)
- [Seaborn](#)

17. Pandas with Excel

- Reading Multiple Sheets
- Writing Formatted Excel
- Excel Automation Basics

18. Pandas with SQL

- Connecting MySQL Database
- Reading SQL Tables
- Writing Data to SQL